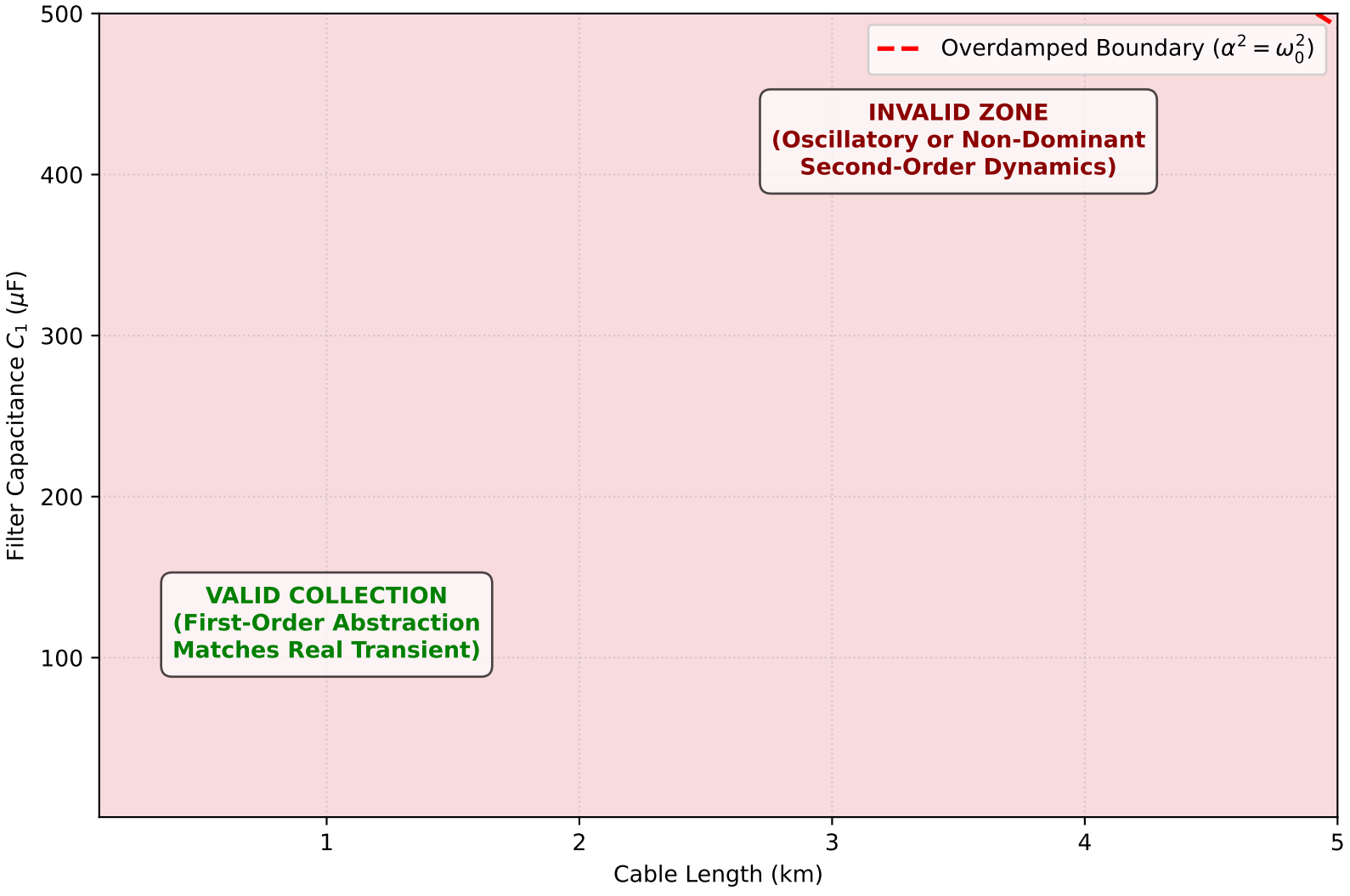
Geometric Boundaries of the "Valid Cables Collection" (M_3 Design Space)
Overdamped Boundary ($\alpha^2 = \omega_0^2$)
INVALID ZONE
(Oscillatory or Non-Dominant
Second-Order Dynamics)
VALID COLLECTION
(First-Order Abstraction
Matches Real Transient)
Filter Capacitance C_1 (μF)
Cable Length (km)
500
400
300
200
100
1
2
3
4
5